Use-Case Flow Specification

Internal Microservice Catalog & Health Portal — Lab 1

Use Case ID: UC-05	**Use Case Name:** Run Health Check Probe
Primary Actor: Scheduler (automated timer)	**Related Use Cases:** «include» UC-06 — Compute Rolling Availability; «extend» UC-07 — Dispatch Downtime Alert

Preconditions

- The microservice is registered in the catalog with a valid, reachable health-check endpoint URL (FR-004).
- The Scheduler service is running and its clock is synchronized with the portal backend.

Postconditions (Success)

- The probe result (success/failure, timestamp, response time) is written to the health log.
- The service's rolling 24-hour availability percentage is recalculated and up to date (FR-001).
- The service's status indicator on the dashboard reflects the latest probe result.

Main Success Scenario

1. The Scheduler triggers a health-check cycle every 30 seconds for each registered microservice.
2. The system sends an HTTP GET/HEAD request to the microservice's configured health endpoint.
3. The microservice responds with a 200 OK status within the timeout window.
4. The system records the probe result (success, timestamp, response time) in the health log.
5. The system executes «include» UC-06 – Compute Rolling Availability, recalculating the 24-hour rolling availability percentage from the updated log.
6. The system updates the service's status indicator to "Healthy" on the dashboard.
7. The use case ends; the Scheduler awaits the next 30-second cycle.

Alternate Flow — A1: Health Probe Fails

Triggered at step 3 when the microservice does not respond within the timeout window or returns a non-2xx status.

The system records the probe result as "Failed" with timestamp and error code.

The system checks the count of consecutive failed probes for this service.

If fewer than 3 consecutive failures: the system sets status to "Degraded" and resumes the main flow at step 1 on the next cycle.

If exactly 3 consecutive failures are reached: the system triggers the extension point «extend» UC-07 – Dispatch Downtime Alert, notifying the on-call DevOps Engineer via the configured channel (e.g., Slack/PagerDuty), and sets status to "Down."

The use case ends (alert dispatched); normal 30-second probing resumes once the service responds successfully again.

Postconditions (Failure Path)

- A downtime alert has been dispatched to the notification channel if the 3-consecutive-failure threshold was reached (satisfies FR-005 acceptance criteria).
- The service's dashboard status accurately reflects "Degraded" or "Down."